Can an AI optimize a growth medium in 24 hours?

Team ViNatX — Monomer Bio / Elnora AI Science Hackathon

Vibrio natriegens

Fastest-growing non-pathogenic bacterium known

~10 min doubling time

→

Monomer Bio Workcell

OT Flex liquid handler
Tecan plate reader
Liconic incubator

Automated OD600 reads

→

Claude Code + MCP

Designs experiments
Submits protocols
Analyzes data
Proposes next round

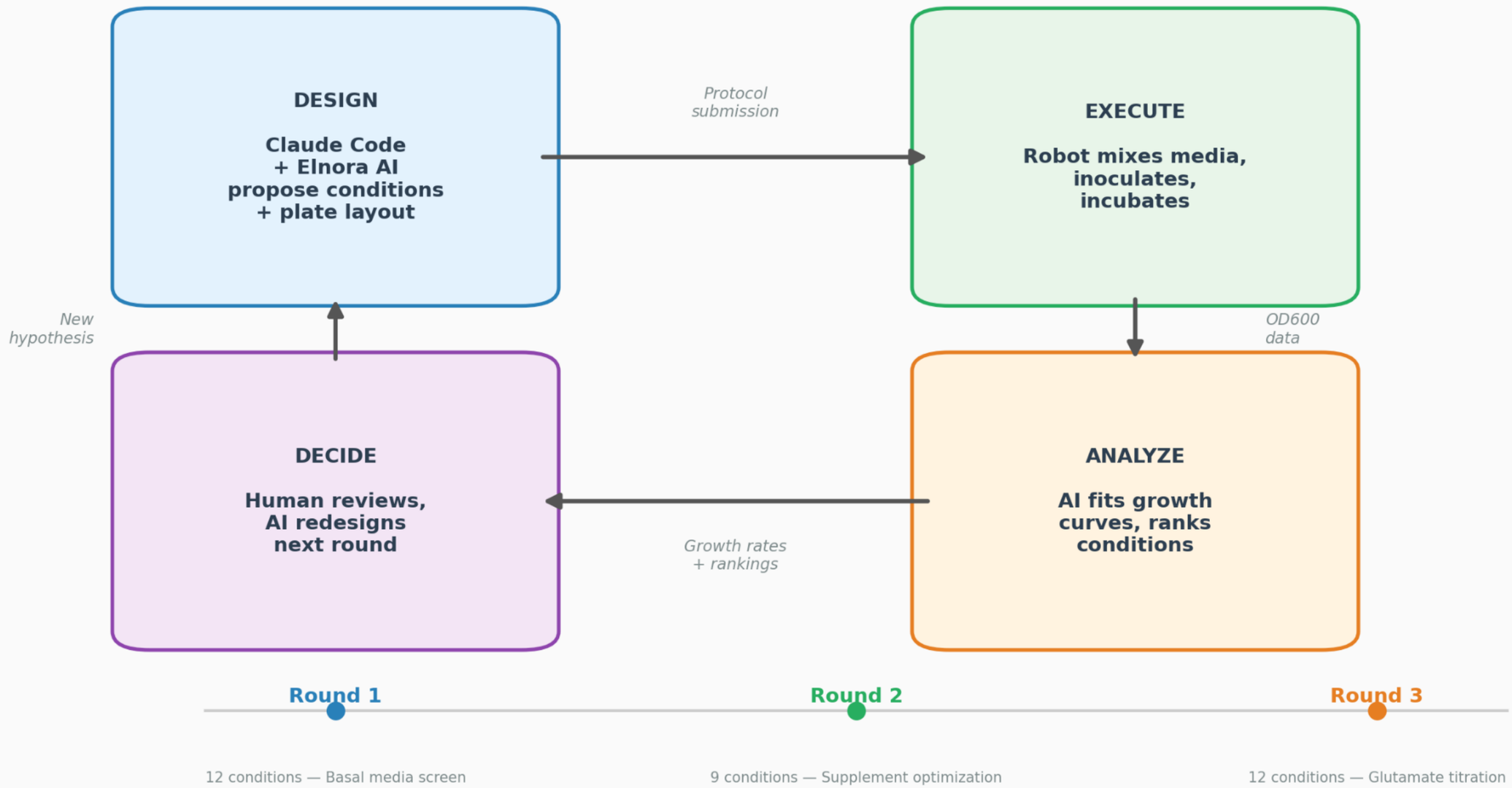
THE GOAL

Starting from 12 candidate media formulations, iterate autonomously to find the fastest-growing condition.

3 rounds · 33 conditions · 1 plate · < 24 hours

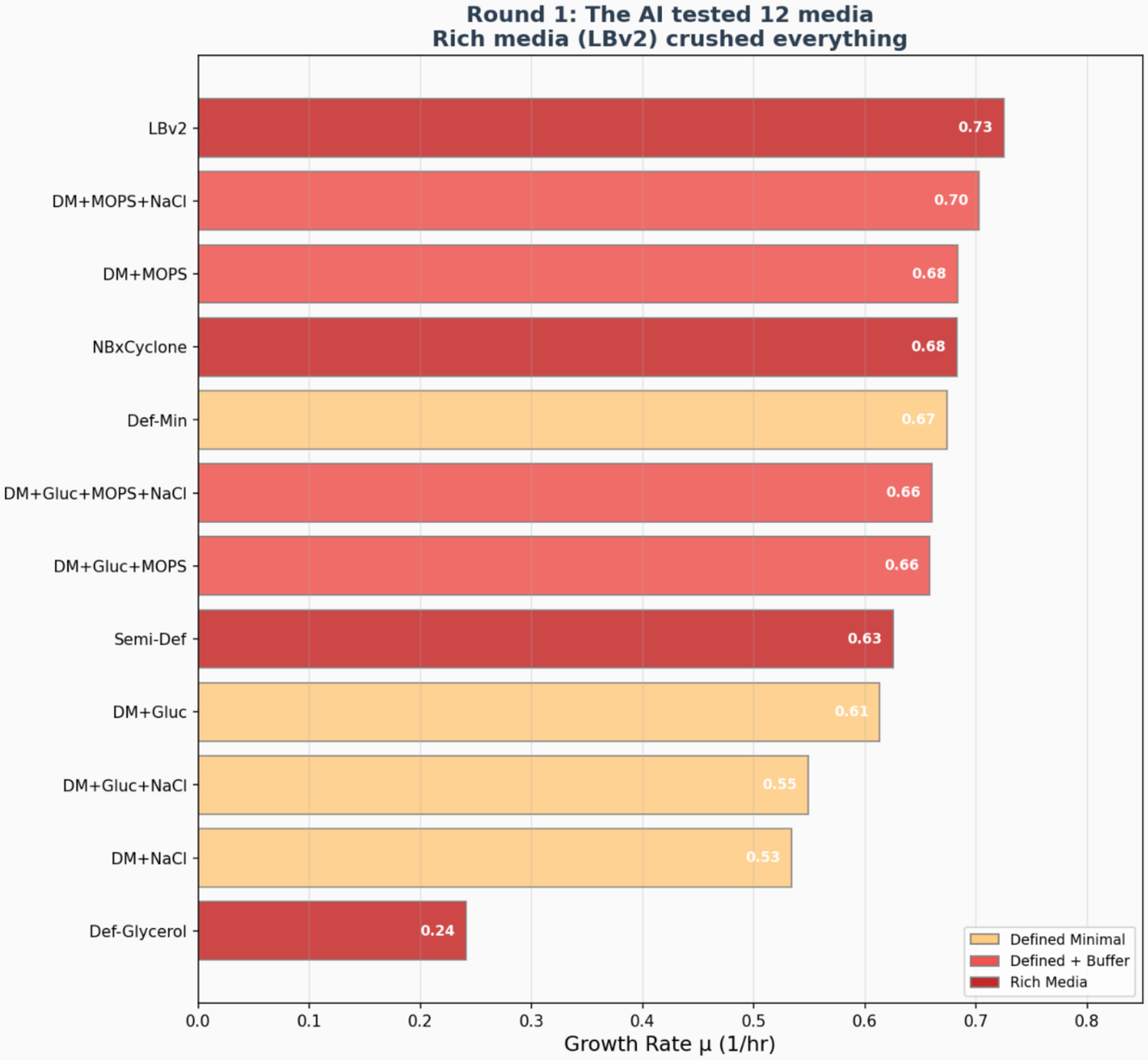
The Autonomous Loop

Design → Execute → Analyze → Repeat — each round in ~2 hours



Wrong Fast, Then Right Fast

The AI's first hypothesis was wrong — but rapid iteration let us pivot



AI'S INITIAL HYPOTHESIS

"Optimize glucose, NaCl, and MOPS buffering on Defined Minimal base medium"

✗ WRONG — defined media were the worst performers

pivot in minutes

DATA-DRIVEN PIVOT

"LBv2 is the winner. Now optimize nitrogen supplementation on LBv2."

✓ Round 2: tested glutamate, tryptone, YE on LBv2

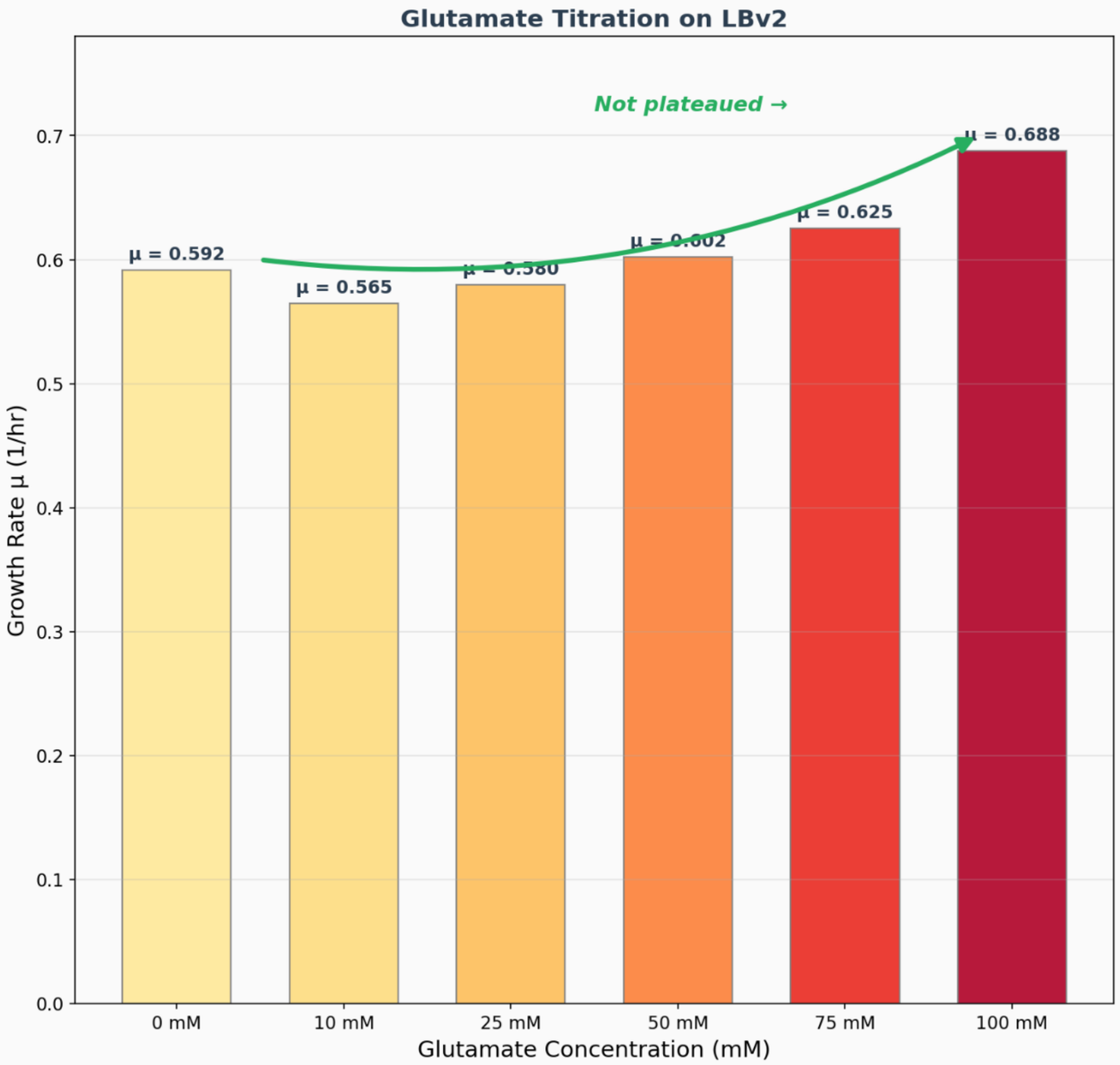
CONFIRMED IN ROUND 3

"Glutamate is the key driver. Clear dose-response: more glutamate = faster growth."

Glucose, tryptone, YE → no help or actively harmful

The Result: A Clear Dose-Response

More glutamate = faster growth — and we haven't hit the ceiling



+16%

growth rate improvement

BEFORE

Bare LBv2
μ = 0.592 /hr

→

AFTER

LBv2 + Glut 100mM
μ = 0.688 /hr

- Glucose actively hurts growth (overflow metabolism)
- Extra tryptone / yeast extract don't help
- Good reproducibility: R2 vs R3 replicate ≈ 3% diff
- Dose-response hasn't plateaued — room to push higher

What We Learned About AI + Biology

The value isn't in getting it right the first time — it's in iterating fast

AI Got the First Hypothesis Wrong

The initial DOE focused on glucose, NaCl, and buffering on Defined Minimal.

The real drivers turned out to be nitrogen source (glutamate) and base media choice (LBv2).

But AI helped us figure that out midstream and pivot in minutes, not days.

Integration Was the Bottleneck

Connecting MCP servers, Notion, Monomer Cloud, and the workcell took real effort up front.

But once the pipeline was built, iteration accelerated dramatically. Rounds 2 and 3 moved much faster than Round 1.

Context consolidation was key.

AI Makes Novel Mistakes (and Bets)

AI made mistakes a grad student would also make: wrong time windows, mismatched comparisons.

But it also lacked the biases that would discourage novel experiments. It proposed conditions we might have vetoed as 'too unusual.'

Some of those could have worked.

3 rounds · 33 conditions · < 24 hours · 16% growth rate improvement · dose-response not plateaued

The right question isn't "Was AI right?" — it's "Did AI help us learn faster?" The answer is yes.